In process

Patrick McQuestion <pmcquest@nd.edu> Thu, Jul 9, 2026 at 5:30 PM
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Hey all, I just wanted to raise a conceptual issue I am having around measuring impacts, and understanding what the LOO analysis can actually tell us. I tried to articulate it in our meeting today before Matt connected, but not very well.

Basically, when we take out a criterion before calculating the subnational democracy index, we can make two subsequent observations:

- (a) the change in the level or "mean impact" this has on the index, essentially by subtracting the partial index score from the full index score, and
- (b) the change in spread or "differential impact" from one municipality-year to the next.

For example, when we take out the Ruling Party (`compete`) criterion from the full index, calculate the partial index, and subtract the full index from the partial index, this can shift our average subnational score up or down, *but it can also shift individual municipalities* up or down in their position relative to other municipalities. In other words, Ruling Party support may be a more influential characteristic for one municipality's expected democracy score than for another's over time. If we find variables that have both mean and differential impacts, we can describe them as more "impactful" than others, and vice versa.

This analogy from Claude also helped me understand the difference: "imagine a company gives every employee a flat $500 raise, while another gives some employees $2,000 and others nothing. Both changes move the average salary by a similar amount (both score high on level), but only the second one changes who out-earns whom (only the second scores high on variation). A metric that only checks 'did anyone's relative position change' (e.g. a correlation between the index with and without a criterion) would call the flat raise close to a non-event, even though the same total amount of money moved."

I also ran a LOO analysis after meeting with Matt this morning to work through this interpretation issue. In my analysis, I find that `compete` (similar support for ruling and non-ruling parties) and `urban` (non-rural municipalities) have the largest average effect on the index *and* the widest spread across municipalities. `prox_cap` (proximity to Bogota) is a split case: third by average effect, but one of the smallest spreads (SD 0.015 vs. 0.030 for `compete/urban`)--nearly every municipality moves by about the same amount when it's removed, regardless of how close or far it actually is from the capital. `econ_dev`--the variable most representative of modernization theory (responding to the question of whether the index is anything more than repackaged modernization theory) is more middling than these variables, and therefore relatively less "impactful."

In the end, the index appears to be quite robust to any characteristic left out, suggesting we have built a decent multi-dimensional measure.

Here are some key metrics and results from my parallel LOO analysis:

Full snvdem Index for Colombia:

Municipalities	1,122
Years	24 (2000-2023)
Municipality-years	26,928
`sndem` (composite index)	mean 0.342, SD 0.068, range [0.121, 0.556]
`snelect` (elections)	mean 0.331, SD 0.071
`sncivlib` (civil liberties)	mean 0.352, SD 0.068

LOO summary table (`loo_sndem - sndem_reconstructed`):

Criterion	Mean impact (sndem)	Median	SD	P05	P95
compete	+0.0237	+0.0239	0.0300	-0.0265	+0.0698
urban	+0.0180	+0.0182	0.0310	-0.0310	+0.0666
prox_cap	+0.0165	+0.0150	0.0150	-0.0062	+0.0430
econ_dev	+0.0125	+0.0129	0.0151	-0.0129	+0.0362
nonviolent	-0.0136	-0.0136	0.0241	-0.0512	+0.0241
nonindig	-0.0126	-0.0101	0.0203	-0.0469	+0.0163
nonremote	-0.0117	-0.0121	0.0169	-0.0397	+0.0154
south	-0.0076	-0.0028	0.0120	-0.0323	+0.0031
pop_density	-0.0075	-0.0075	0.0127	-0.0268	+0.0121
east	-0.0044	-0.0021	0.0073	-0.0169	+0.0008
west	-0.0032	-0.0024	0.0077	-0.0184	+0.0062
north	-0.0023	-0.0013	0.0054	-0.0138	+0.0043

What a mean impact of +0.0237 (`compete`) actually looks like in the data, using a cross-section for 2023:

- A municipality (DANE code 19845: Villa Rica, Cauca) with a full-index score of 0.409 becomes 0.502 without `compete` (+0.093, one of the largest single moves that year); a municipality (94884: Puerto Colombia, Guainia) with a full-index score of 0.240 becomes 0.174 (-0.066, one of the largest drops). A third (88001: San Andrés) moves from 0.382 to 0.405--almost exactly the +0.0237 mean.
- 76.5% of municipality-years move up when `compete` is excluded, 23.5% move down.
- In SD terms, +0.0237 is about a third of sndem's own cross-sectional spread (SD 0.068). For a municipality sitting exactly at the median, a shift that size moves it from the 50th percentile to roughly the 62nd-63rd percentile of the full distribution. Nobody jumps from bottom decile to top decile on the strength of one criterion, for example, so it's not that dramatic in the end.
- Splitting the two dimensions: `compete`'s effect is similar in size on both (Mean_EMEL +0.026, Mean_CSCW +0.022), not concentrated in elections or civil liberties specifically.
- *Does the relative ranking of municipalities change, not just the average?* Yes. Within 2023 alone, the Spearman rank correlation between the full and `compete`-excluded ranking is 0.96--high, as expected for dropping 1 of 12 terms--but the average municipality still moves 68 rank positions (out of 1,122), and 581 of 1,122 municipalities (52%) move by more than 50 positions. A correlation of 0.96 and high reshuffling are both true at once.

Here are my thoughts on Matthew's points, comparing his and my results:

- *Still good results, all characters are correlated above .9. Nothing collapses the index when we remove it.*
  - True, but I believe this correlation only tracks whether municipalities' relative ranking survives, not how much actually moved. It's close to a guaranteed outcome when dropping 1 of 12 comparably-weighted terms for the full panel dataset.
- *Rural/Urban and Ruling Party Strength now are the most impactful characteristics here. Removing either does affect the model pretty heavily.*
  - Confirmed; they're my top 2 by level-shift too.
- *"South" is now negative in its weighting.*
  - I think this result has to do with non-independence between characteristics and spatial clustering in our data. The municipalities scoring highest on "south" are almost entirely in the provinces/departments of Amazonas, Putumayo, Vaupés, Caquetá, Nariño, Guaviare, Guainía. Those same departments are the most remote, most Indigenous-population-dense, and least developed in the data. When I separate out the effect of those three factors, `south`'s effect actually turns positive, so the negative number we're seeing could actually be remoteness, Indigenous population, and development showing up.
- *Remoteness and Population Density are now the most explanatory, above economic development and rural/urban.*
  - Agreed. This is regarding the variance decomposition section in Matt's script (lines 152-203), which asks: "looking at just this one criterion's own score, on its own, how well does it predict where a given municipality lands on the final, all-12-criteria index?" Remote municipalities in this data tend to score in similar ways on several other criteria (development, historically weaker state presence, etc.), so remoteness ends up tracking the whole composite closely even though its own individual contribution to any single shift

    - is modest. `compete` is the mirror image: it carries enough weight to move the average significantly more when removed, but its own pattern across municipalities doesn't line up with the composite as tightly.
- *Distance to the capitol is still in the middle.*
    - This is the clearest illustration of the two-meanings-of-"impactful" question above. `prox_cap` (proximity to Bogota) has one of the largest average effects in my analysis (rank 3) but one of the smallest spreads across municipalities; it moves almost every municipality by about the same amount, near-capital or far. Both rankings are correct; they're answering different questions.

I thought it would be an opportune moment to raise the conceptual conversation before deciding where it fits in our manuscript. I think the LOO analysis would make for a useful appendix, and we can refer readers to it when describing our results. Thanks for reading,
-Patrick

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